8 20 men, 10 women and 10 children are in a competition.

The mean score for the women is 15.6
The mean score for the children is 9.2

Kevin says that the mean score for all 40 people is 11.2

(a) Work out the mean score for the men.

Women total:15.6 × 10 = 156.

Children total:9.2 × 10 = 92.

All40 total:11.2 × 40 = 448.

Men total:448 – 156 – 92 = 200.

Men mean:200 ÷ 20 = 10.

1 mark
Mean × frequency

1 mark
Men remainder ÷20

1 mark
10

(3)

Kevin was wrong.
The mean score for all 40 people was actually 11.15

(b) How does this affect the mean score for the men?

Reduced:the total score falls but20 men remain.

New mean = (11.15 × 40 – 156 – 92) ÷ 20 = 9.9.

1 mark
Mean decreases

(1)

(Total for Question 8 is 4 marks)